Approximately Kinematic Waveform Inversion Methods

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Agenda

Waves & Rays

Cycle-skipping

Adaptive Waveform Inversion

Not done yet...

Acoustic waves radiating from isotropic point source at $\mathbf{x}_s \in \mathbf{R}^3$ with waveform $f(t)$ - solves

$$\Box q(\mathbf{x},t;\mathbf{x}_s) = \left(\frac{1}{\kappa}\frac{\partial^2 q}{\partial t^2} - \frac{1}{\rho}\nabla^2 q\right)(\mathbf{x},t;\mathbf{x}_s) = f(t)\delta(\mathbf{x}-\mathbf{x}_s),\ q = 0 \text{ for } t \ll 0$$

$q =$ potential
$\kappa =$ bulk modulus
$\rho =$ density
$f =$ source time function
$p = \rho\partial_t q =$ pressure

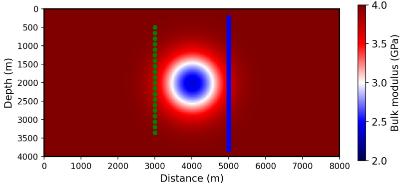
Waveform data:: emit waves at $\mathbf{x}_s$, record pressure at $\mathbf{x}_r$ Source-receiver pairs $X_{sr} = \{(\mathbf{x}_s, \mathbf{x}_r)\}$

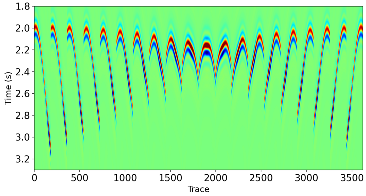
Simplify: density $\rho > 0$, source waveform $f \in C_0^\infty(\mathbf{R})$ known

Predicted pressure data $p[\kappa](\mathbf{x}_r, t; \mathbf{x}_s)$

Dynamic Inverse Problem: given data $\{d(\mathbf{x}_r, \cdot; \mathbf{x}_s) : (\mathbf{x}_s, \mathbf{x}_r) \in X_{sr}\}$, seek $\log \kappa \in C^\infty(\mathbf{R}^3)$ so that $p[\kappa] \approx d$.

Many variants, large literature since '70's, many apps





Left: 2D bulk modulus field ("lens") κ^*, 20 source positions ($\mathbf{x}_s = (x_s, z_s)$, green) at $x_s = 3000$ m, 181 receiver positions ($\mathbf{x}_r = (x_r, z_r)$, blue) at $x_r = 5000$ m. Right: predicted data $d^*(\mathbf{x}_r, t; \mathbf{x}_s) = p[\kappa^*](\mathbf{x}_r, t; \mathbf{x}_s)$. (from S., Chen & Minkoff, *Inverse Problems* 41 (6), 2025)

High frequency asymptotics (geometric optics/acoustics,...): in HF limit, travel time of wave from source at $\mathbf{x}_s$ to receiver at $\mathbf{x}_r$ is *geodesic distance* $\tau[\kappa](\mathbf{x}_s, \mathbf{x}_r)$ in metric $c(\mathbf{x})^{-2}\sum_i dx_i \otimes dx_i$, $c = \sqrt{\kappa/\rho}$ = sound speed

Kinematic Inverse Problem: given times $\{t(\mathbf{x}_s, \mathbf{x}_r) : (\mathbf{x}_s, \mathbf{x}_r) \in X_{sr}\}$, find κ so that $\tau[\kappa] \approx t$

Huge literature - eg. boundary rigidity problem (find c in $\Omega \subset \mathbf{R}^n, X_{sr} \subset \partial\Omega$), many apps (seismic tomography,...)

Complication: multiple geodesics ("rays") may connect source/receiver points - which τ to match?

Common computational approach: pose as optimization, eg. least squares - use iterative algorithm to find κ to (approximately) minimize

$$J_{\mathrm{FWI}}[\kappa, d] = \sum_{(\mathbf{x}_s, \mathbf{x}_r) \in X_{sr}} \int dt \, |p[\kappa](\mathbf{x}_r, t; \mathbf{x}_s) - d(\mathbf{x}_r, t; \mathbf{x}_s)|^2$$

("Full Waveform Inversion") or

$$J_{\mathrm{TT}}[\kappa, t] = \sum_{(\mathbf{x}_s, \mathbf{x}_r) \in X_{sr}} |\tau[\kappa](\mathbf{x}_s, \mathbf{x}_r) - t(\mathbf{x}_s, \mathbf{x}_r)|^2$$

("Traveltime Tomography")

$+$ regularizations, constraints

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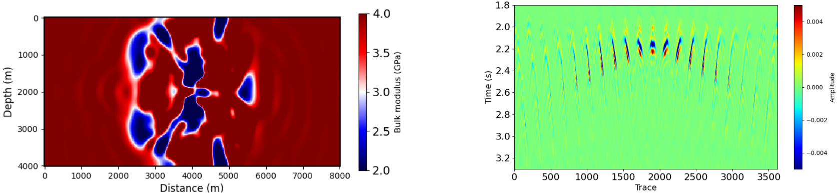
Good news, bad news...

For FWI,

- seismology, ultrasonic imaging, ocean acoustics,... the data are pressure (or acceleration, or stress, or ...), so FWI is the natural approach, BUT...
- unless initial κ predicts traveltimes accurately ("within a half-wavelength"), local optimization (eg. quasi-Newton), required because frequency range $\Rightarrow$ fine discretization $\Rightarrow$ large problem size, tend to stagnate at incorrect final κ with big data prediction error - wavelength-related ("cycle-skipping")



Cycle-skipping in action. Left: Estimated κ_{200} from 200 LBFGS iterations on J_{FWI}, from initial estimate $\kappa_0 \equiv 4.0$ GPa, using data d from "lens" κ. Right: residual $p[\kappa_{200}] - d$. RMS relative error $= 0.43$.

For TT,

- J_{TT} *a priori* unrelated to wavelength, optimizations tend to converge to good fit, BUT...
- data t are not directly measured, must be inferred ("picked") from waveform data - mostly automated in industrial codes, but can be noise-sensitive

Conventional remedy for cycle skipping: pick times $t(\mathbf{x}_r, \mathbf{x}_s)$, minimize $J_{\rm TT}$, initiate minimization of $J_{\rm FWI}$ from result, low $\rightarrow$ high frequency continuation - widespread use, much success (including commercial)

Still, many attempts at alternatives: is there a replacement for FWI that uses waveform data directly but does not cycle-skip, implicitly accomplishes travel-time inversion?

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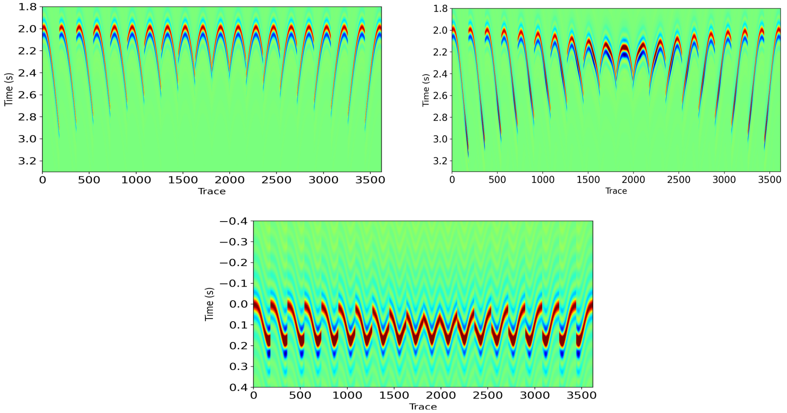
Warner et al. (2012, ...): FWI fails because initial predicted data $p[k]$ places waves at wrong times $\Rightarrow$ big error - so *move them* to make error small

Use a *filter* (convolution kernel) $u_\sigma[\kappa, d](\mathbf{x}_r, t; \mathbf{x}_s)$, solution of

$$u_\sigma[\kappa, d] = \mathrm{argmin}_u(\|p[\kappa] * u - d\|^2 + \sigma\|u\|^2)$$

regularization weight $\sigma > 0$ TBD

($* =$ convolution in t, $\|\cdot\| = L^2$ norm)



Top Left: $p[\kappa_0], \kappa_0 \equiv 4.0$ GPa. Top Right: target data $d = p[\kappa^*]$. Bottom: filter $u_\sigma[\kappa_0, d]$, achieves $p[\kappa_0] * u_\sigma \approx d$ (least squares, via CG iteration)

BUT desiired outcome is $p[\kappa] \approx d$ which is $p[\kappa] * u \approx d$ with $u = \delta(t)$

Adaptive Waveform Inversion: Since $t\delta(t) = 0$, try minimizing (over κ)

$$J_{\mathrm{AWI},\sigma}[\kappa, d] = \sum_{(\mathbf{x}_r, \mathbf{x}_s) \in X_{sr}} \frac{\|tu_\sigma[\kappa, d](\mathbf{x}_r, \cdot; \mathbf{x}_s)\|^2}{\|u_\sigma[\kappa, d](\mathbf{x}_r, \cdot; \mathbf{x}_s)\|^2}$$

Many successful results reported - overcomes cycle skipping! (at least sometimes...)

Why should this work?

Under some conditions $J_{\mathrm{AWI},\sigma}[\kappa, d] \rightarrow J_{TT}[\kappa, t]$ as data wavelength $\rightarrow 0$

Assume:

only one geodesic connects points near sources and receivers (simple metric);

once a geodesic leaves neighborhood of X_{sr}, it never returns;

exponential time decay of solutions (eg. non-trapping metric, Vainberg '75);

asymptotics via family of source waveforms $\{f_\lambda(t) = \sqrt{\lambda} f_1(t/\lambda)\}$, $f_1 \in \partial_t^2 C_0^\infty(\mathbf{R})$, corresponding pressures $p_\lambda[\kappa]$.

noise-free data $d^*_\lambda = p_\lambda[\kappa^*]$.

Then:

$$\|p_\lambda[\kappa] * u_\sigma[\kappa, d^*_\lambda] - d^*_\lambda\| \ll \|d^*_\lambda\| \Leftrightarrow \sigma \approx \text{ const. } \lambda$$

so assume $\sigma \propto \lambda$ as $\lambda \rightarrow 0$, then:

$u_\sigma[\kappa, d^*_\lambda], tu_\sigma[\kappa, d^*_\lambda] \in L^2(\mathbf{R} \times X_{sr})$, so $J_{\mathrm{AWI},\sigma}$ is well-defined, and for sufficiently small σ (and λ),

$$|J_{\mathrm{AWI},\sigma}[\kappa, d^*_\lambda] - J_{\mathrm{TT}}(\kappa, t^*)| \leq C\lambda^2$$

$(t^*(\mathbf{x}_s, \mathbf{x}_r) = \tau[\kappa^*](\mathbf{x}_s, \mathbf{x}_r))$

Sketch of proof:

Hadamard expansion of Green's function (3D) $\Rightarrow$ for solution of $\Box q = f(t)\delta(\mathbf{x} - \mathbf{x}_s)$ in simple metric domain (unique geodesic connects $\mathbf{x}_s$ to $\mathbf{x}$):

$$q(\mathbf{x}, t; \mathbf{x}_s) = a(\mathbf{x}.\mathbf{x}_s) f(t - \tau(\mathbf{x}, \mathbf{x}_s)) + \int_{\tau(\mathbf{x},\mathbf{x}_s)}^{t} ds\, b(\mathbf{x}, s; \mathbf{x}_s) f(t - s)$$

a, b smooth for $\mathbf{x} \neq \mathbf{x}_s$ - "No Return" $\Rightarrow$ global in t

So $p[\kappa](\mathbf{x}_r, t; \mathbf{x}_s) = \rho a(\mathbf{x}_r, \mathbf{x}_s) \partial_t f(t - \tau(\mathbf{x}_r, \mathbf{x}_s)) + ...$

(Hadamard 1923, Courant & Hilbert '62, Friedlander '75, Hörmander '85, Wei et al. '24) ignore "...", noise free data $d^* = p[\kappa^*] \approx \rho a^* f(t - \tau^*)$ ($a^* = a[\kappa^*]$ etc.) so

$$u_\sigma = \mathrm{argmin}_u(\rho^2 \| a\partial_t f * u(t-\tau) - a^*\partial_t f(t-\tau^*)\|^2 + \sigma\|u\|^2)$$

$$= \frac{a^*}{a} g_\sigma(t - (\tau - \tau^*)),\ \hat{g}_\sigma(\omega) = \frac{\rho^2 a^2 \omega^2 |\hat{f}(\omega)|^2}{\rho^2 a^2 \omega^2 |\hat{f}(\omega)|^2 + \sigma}$$

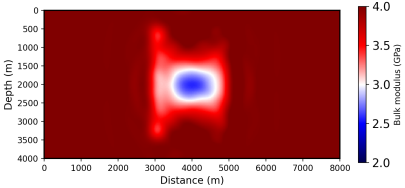
whence

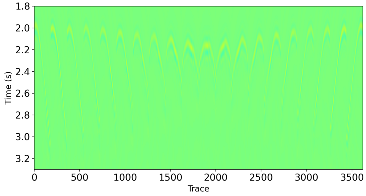
$$J_{\mathrm{AWI},\sigma} \approx \sum_{(\mathbf{x}_r,\mathbf{x}_s)\in X_{sr}} \left((\tau - \tau^*)^2 + \frac{\|tg_\sigma\|^2}{\|g_\sigma\|^2} \right)$$

choose $f = f_\lambda$ and $\sigma \propto \lambda$ then

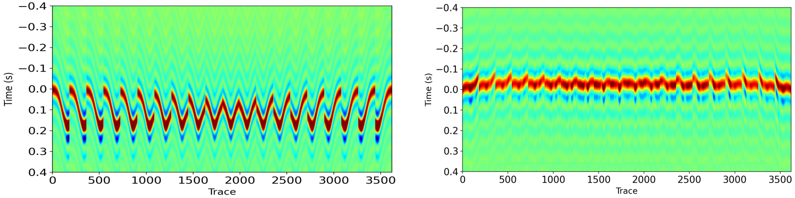
$$= \|\tau - \tau^*\|^2 + O(\lambda^2)$$

$O(\lambda^2)$ estimates for "...", other fine print - S. '24 (arXiv) (related: Song '94)





MSWI ($=$ AWI w/o denominator, just $\|tu_\sigma\|^2$, similar local asymptotics) $+$ FWI: 12 LBFGS iterations each. Left: final estimate of κ. Right: final residual, plotted on same scale as data, 7% RMS error.



Filter u_σ. Left: at initial estimate $\kappa \equiv 4.0 GPa$. Right: at MSWI estimate of κ, 12 LBFGS iterations (initial estimate for FWI).

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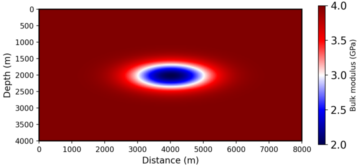
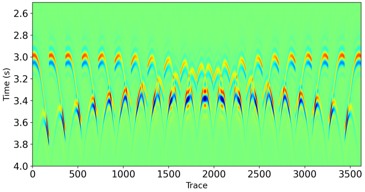
Metric not simple (multiple geodesics connect source, receiver positions) $\Rightarrow$ asymptotic relation AWI $\leftrightarrow$ TT fails





Left: extended "lens". Right: data traces, sources at $x_s = 2000$ m, receivers at $x_r = 6000$ m.

Interfernce between multiple waves (travel times) $\Rightarrow$ AWI behavies just like FWI ("cycle skips") (S. '94, Huang 19, Pladys '21, Yong '23, S. '24)

Modifications of AWI to deal with non-simple metrics:

- Localize in time: Local AWI (Yong '23.) arriving waves (usually) separated in time - localize via Gabor transform
- Microlocalize in space-time: dense source/receiver arrays, spatially extended virtual sources ("acoustic antennae"), penalize extension in space (Huang '19, S. '23) - asymptotic to non-standard traveltime misfit ($\sim$ lens rigidity), also in presence of multiple travel times

Completely untouched but extremely important: elasticity, EM, reflection (non-smooth coefficients)

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